Luca Daloia

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EDUCATION:

North Carolina State University, Raleigh, NC – Expected Dec 2026
Dual BS: Electrical & Computer Engineering - Cum. GPA 3.39 (Dean's list 2022, 2024, 2025, Honors Program invitation)

WORK EXPERIENCE:

Summer 2025 - NC State Univ (Research Support for Prof. John Muth, PhD - Progress Energy Distinguished Prof)
Built a Django web app to organize, visualize, compare & benchmark semiconductor devices from research publications.

Summer 2024 – Corning Inc. (Division Controls Engineer Intern)
Built Python tools for automated extruder monitoring, data analysis and management. Authored technical documentation.

Summer 2023 – Daloia Consulting (Personal Project)
Developed a Python Tkinter GUI with SQLite backend database to compute minimum conduit sizes per National Electrical Code requirements, storing user inputs in an editable database and streamlining calculations to reduce time and errors.

RELEVANT COURSEWORK & HANDS-ON PROJECTS

Senior Design - Auger Alignment System – Sponsored by Caterpillar Inc (Awarded Best-in-Category): designed a multilayer KiCad PCB integrating ESP32s with high-precision IMU and power management; architected the full hardware solution (from schematic capture through PCB layout) and developed C firmware to process inertial data and provide real-time orientation via ESP-Now and an LED interface for precise field alignment. The project included prototyping, system integration, hardware and software testing (including embedded-system validation), and iterative fixes to mature design.

Intro. to Embedded Systems: Designed bare-metal MSP430 autonomous robot car with power regulation, H-bridge motor control, and multi-sensor fusion. Programmed the embedded system firmware in C to process ADC data from IR sensors for boundary detection and switch between autonomous line-following and manual control, and implemented a PID controller for the autonomous line-following logic to produce smooth, stable corrective steering. Used PWM and ADC/DAC for motor control and inputs. Integrated a Wi-Fi module (TCP/IP) and built a remote control steering interface for a full-stack IoT solution using UART for serial communication between IoT module and MSP430. Verified hardware with an Analog Discovery (AD3), Digital MultiMeter (DMM), and oscilloscope, and debugged in Code Composer Studio.

Special Topics – Embedded Systems Hardware Design: Engineered a TI MSP430 microcontroller-based embedded system by interpreting schematics to design the power circuit validated through LTSpice simulations. Using KiCad, I developed a multi-layer PCB integrating LDRs, LEDs, and tactile buttons while managing complex routing, power planes, and mechanical constraints. Developed C-firmware to implement system functionality.

Special Topics – Intro. to Robotics & Autonomous Systems: Robotic fundamentals (dynamics, feedback control, motion planning, trajectory optimization, state estimation, localization, perception, robot learning) and system-level understanding of modern autonomous systems ranging from self-driving cars to humanoid robots.

Fund. of Logic Design: Built a state machine-based circuit using a 555 timer, D flip-flops, and integrated circuit chips to implement sequential LED turn signals in a model car. Designed and optimized it with Boolean simplification and K-Maps using Verilog to develop testbenches to validate circuit integrity, and integrated the circuit into the prototype car.

Elem. of Control Systems: Simulated a MATLAB self-balancing Segway: derived dynamics from DC motor and inverted pendulum models, created transfer-function & state-space representations, built full-state stabilization feedback controller.

SKILLS

- Languages: Python (Django, Tkinter, SQLite), C/C++, Assembly, LaTeX
- Coding Tools & Version Control: VS Code, Antigravity IDE, Git, GitHub, Code Composer Studio (CCS) IDE & Debugger
- HW & Embedded Systems: HDL (Verilog), Embedded systems (ESP32, MSP430), Schematic capture & PCB design, Soldering, LaTeX (documentation)
- EDA/Simulation & Modeling: KiCad (schematic & PCB layout), MATLAB, LTspice
- 3D Design and Printing: Autodesk Fusion 360
- Test, Measurement & Workshop Tools: Analog Discovery3/WaveForms (logic analyzer, scope, generators), Oscilloscope, Digital multimeter (DMM), Bench power supply, Wireshark, Dremel, Drill press
- Office: MS Windows and Kali Linux, MS Office (Outlook, OneNote, Teams, OneDrive, Word, PowerPoint, Excel - Pivot tables, macros, functions), Google Workspace (gmail, Meet, Docs, Sheet, Slides, Drive), Dropbox, Oracle VirtualBox
- AI Assistants: Effective and efficient use of Assistants (ChatGPT,Gemini,Copilot) while careful with sensitive information
- Soft skills: strong analytical and problem-solving, clear verbal and written communication, collaborative team player, leadership, responsibility, curiosity, creativity, initiative, adaptability, emotional intelligence, respect, professionalism

PROJECT PORTOFOLIO   **https://github.com/lucadaloia**